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Genome-wide DNA methylation and mRNA transcription analysis revealed aberrant gene regulation pathways in patients with dermatomyositis and polymyositis

Hui Luo[1,2,3], Honglin Zhu[1,2,3], Ding Bao[1,2,3], Yizhi Xiao[1,2,3], Bin Zhou[1,2,3], Gong Xiao[1,2,3], Lihua Zhang[1,2,3], Siming Gao[1,2,3], Liya Li[1,2,3], Yangtengyu Liu[1,2,3], Di Liu[1,2,3], Junjiao Wu[1,2,3], Qiming Meng[1,2,3], Meng Meng[1,2,3], Tao Chen[4], Xiaoxia Zuo[1,2,3], Quanzhen Li[1,4], Huali Zhang[1,5]

[1]Department of Rheumatology and Immunology, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China;
[2]National Clinical Research Center for Geriatric Disorders, Xiangya Hospital, Central South University, Changsha, Hunan 410008, China;
[3]Provincial Clinical Research Center for Rheumatic and Immunologic Diseases, Xiangya Hospital, Changsha, Hunan 410008, China;
[4]Microarray and Immune Phenotyping Core Facility, University of Texas Southwestern Medical Center, Dallas, TX, USA;
[5]Department of Pathophysiology, Xiangya School of Medicine, Central South University, Changsha, Hunan 410008, China.

To the Editor: Idiopathic inflammatory myopathies (IIMs) are a diverse group of rare and refractory systemic autoimmune diseases. DNA methylation, influenced by both inherited DNA sequences and environmental exposure, is a critical epigenetic factor that regulates gene expression by modifying the transcriptional chromatin accessibility of regulatory sequences within genes.[1–3] To explore potential pathogenic mechanisms and identify accessible biomarkers for dermatomyositis (DM) and polymyositis (PM), we conducted a comprehensive analysis of alterations in both the DNA methylome and transcriptome in peripheral blood from patients with DM and PM. Additionally, we examined the serum profiles of inflammatory cytokines/chemokines.

The human experimentation protocol for this study received approval from the Ethics Committee of Xiangya Hospital (No. 201212074). Written informed consent was obtained from all participants. Blood samples were collected from a total of 16 patients with DM and 8 with PM who were recruited from the Department of Rheumatology and Immunology at Xiangya Hospital, Central South University, between January 2012 and December 2013. Additionally, 12 age- and sex-matched normal controls (NC) were included in the study. The clinical features are detailed in Supplementary Table 1, http://links.lww.com/CM9/C189.

To characterize DNA methylation and transcriptome alterations in DM and PM patients, genomic DNA and RNA were isolated from peripheral blood mononuclear cells. Besides, multiplex cytokine assays were also conducted using the Bio-Plex Pro Human Cytokine 27-plex Assay (Bio-Rad, Hercules, CA), which contains beads conjugated with monoclonal antibodies specific for interleukin-1 receptor antagonist (IL-1RA), IL-1β, IL-2, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-12p70, IL-13, IL-17A, IL-15, Eotaxin, interferon (IFN)-γ, IFN-γ-inducible protein 10 (IP-10), monocyte chemoattractant protein 1 (MCP-1), macrophage inflammatory protein 1 alpha (MIP-1α), MIP-1β, platelet-derived growth factor-BB (PDGF-BB), granulocyte colonystimulating factor (G-CSF), basic fibroblast growth factor (bFGF), regulated on activation, normal T cell expressed and secreted (RANTES), vascular endothelial growth factor (VEGF), granulocyte-macrophage colony-stimulating factor (GM-CSF), and tumor necrosis factor alpha (TNF-α). Finally, genome-wide DNA methylation analysis, transcriptional profiling, integration of methylation with transcriptional data, and functional classification/gene network analyses were performed. Validation of differential gene and methylation site expression was conducted using real-time PCR, bisulfite pyrosequencing and immunofluorescence. All statistical analyses, excluding those for microarray data, were performed using GraphPad Prism 9 (GraphPad Software, San Diego, California, USA), with a significance threshold set at $P < 0.05$. Detailed information for all methods is provided in the Supplemental Materials, http://links.lww.com/CM9/C189.

The DNA methylation profiles showed separate clusters, indicating a substantial diversity in DNA methylation

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Hui Luo and Honglin Zhu contributed equally to this work.

Correspondence to: Quanzhen Li, Microarray and Immune Phenotyping Core Facility, University of Texas Southwestern Medical Center, Dallas, TX, USA
E-Mail: quanzhenli2013@gmail.com;
Huali Zhang, Department of Pathophysiology, Xiangya School of Medicine, Central South University, Changsha, Hunan 410008, China
E-Mail: zhanghuali@csu.edu.cn

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profiles between DM and NC, as well as PM and NC. In DM patients, 241 differentially methylated cytosine-phosphate-guanine (CpG) sites were identified, the majority of which were hypomethylated (167 CpG sites). Among them, 0% hypomethylated sites were in CpG islands and 23.3% located at gene promoter regions; 14.8% hypermethylated sites were in CpG islands and 25.6% located at gene promoter regions [Supplementary Figure 1A–D, http://links.lww.com/CM9/C189]. A total of 500 differentially methylated CpG sites were identified in PM compared with NC, the majority of which were hypomethylated (347 CpG sites). Among them, 1.1% hypomethylated sites were in CpG islands and 23.9% were located at gene promoter regions; 15.6% of hypermethylated sites were in CpG islands and 30.7% were located at gene promoter regions [Supplementary Figure 1E–H, http://links.lww.com/CM9/C189]. There were 205 common differentially methylated CpG sites in DM and PM, corresponding to 151 hypomethylated and 54 hypermethylated sites [Supplementary Figure 1I–J, http://links.lww.com/CM9/C189].

When compared with NC, a total of 540 differentially expressed genes (DEGs) were identified in patients with DM, including 277 upregulated and 263 downregulated genes, while in patients with PM, a total of 242 DEGs were identified, including 181 upregulated and 61 downregulated; and there were 166 common DEGs in DM and PM [Supplementary Figure 2A–D, http://links.lww.com/CM9/C189]. Gene Ontology (GO) analysis of upregulated genes in DM found those genes mainly enriched in the type I interferon signaling pathway and pathways involved in immune response. Pathway analysis revealed that the significantly pathways included antigen processing and presentation, leukocyte transendothelial migration, and others [Supplementary Figure 2E–G, http://links.lww.com/CM9/C189]. In PM, GO analysis showed that the upregulated genes were mainly involved in innate immune response and inflammatory response. Pathway analysis enrichment revealed the most significant pathways included starch and sucrose metabolism, cytokine–cytokine receptor interaction, and others [Supplementary Figure 2H–J, http://links.lww.com/CM9/C189].

To explore the association between aberrant methylation and differentially expressed genes in DM and PM patients, we mapped the averaged normalized DNA methylation data of the CpG probes to the nearest data and found a significant inverse correlation between methylations and gene expression, with correlation coefficients of –0.0697 for DM and –0.0656 for PM [Supplementary Figure 3A, B, http://links.lww.com/CM9/C189]. We identified DNA methylation alterations in 348 out of 540 differentially DEGs (64.44%) in the DM samples, as well as in 164 out of 242 DEGs (74.38%) in the PM samples [Figure 1A, B]. To pinpoint the most differentially expressed genes, we applied a 0.2 cutoff to the delta-beta values. This analysis revealed that 4 genes (*CXCR6*, *ABLIM1*, *LAX1*, and *PRKCH*) were hypermethylated and downregulated, while 3 genes (*DYSF*, *IL1R2*, and *MYB*) were hypomethylated and upregulated in the DM samples [Figure 1C]. In the PM samples, 4 genes (*CXCR6*, *ABLIM1, CD3G*, and *SLFN5*) were hypermethylated and downregulated, while 6 genes (*OLFM4*, *DYSF*, *IL1R2*, *ARG1*, *PPARG*, and *PDE6H*) were hypomethylated and upregulated [Figure 1D]. The expression levels of four genes—*CXCR6*, *ABLIM1*, *DYSF*, and *IL1R2*—overlapped in both the DM and PM samples. To validate the microarray findings, we conducted RT-PCR to assess the mRNA levels of the 13 genes. Except for *PDE6H* and *MYB*, aboved-mentioned genes exhibited changes identical to those observed in the microarray findings [Figure 1E]. Moreover, *ABLIM1*, *MYB,* and *SLFN5* were expressed at higher levels in IIM patients with interstitial lung disease (ILD) compared to those without ILD [Figure 1F]. Bisulfite-modified DNA pyrosequencing further confirmed the methylation status of 9 genes in both DM and PM patients [Figure 1G]. An intriguing target, *DYSF*, was identified, showing significant upregulation and hypomethylation in both DM and PM patients. *DYSF* encodes the protein dysferlin, a skeletal muscle protein associated with the sarcolemma. We investigated the expression level of dysferlin in muscle tissues using immunofluorescence, and found that dysferlin was increased in the muscle tissues of DM and PM patients [Figure 1H].

IFN is one of the most widely studied cytokine in IIM. Here, we found that 36 *IFN*-related genes were upregulated in DM, 20 *IFN*-related genes were upregulated in PM, and 18 *IFN*-related genes were increased in both [Supplementary Figure 4A, http://links.lww.com/CM9/C189]. We further investigated the methylation status of the CpG sites on the upregulated *IFN* genes. Among the 233 CpG sites in the 18 *IFN*-related genes, 37 CpG sites exhibited hypomethylation in both DM and PM patients compared with NC, including 1 site on six genes (*IFI6, IFIT1, IFT20, OAS1, OAS3, LGALS3BP*), 2 sites on four genes (*VAMP5, HSPA1A, STAT3,* and *IFITM2*), 4 sites on *IFITM3, IGFBP2* and *IGFBP7*, 5 sites on *SERPING1* (serine protease inhibitor (serpin) family G, member 1), and 6 sites on *IFNGR2* [Supplementary Figure 4B, http://links.lww.com/CM9/C189]. The mRNA expression of *SERPING1* and *IFITM3*, along with five CpG sites on *SERPING1* and four CpG sites on *IFITM3*, are displayed in Supplementary Figure 4C and D, http://links.lww.com/CM9/C189.

The quantitative analysis of 27 cytokines/chemokines in the sera of DM, PM, and NC showed that IL-5 and IP-10 were significantly increased, while IL-8 and PDGF-BB were significantly decreased in both DM and PM when compared with NC [Supplementary Table 2, http://links.lww.com/CM9/C189]. At the transcription level, we found that IL-5, CXCL10, and CCL2 were elevated in DM, and PDGFB was decreased both in DM and PM, which is consistent with the serum cytokines/chemokines concentration [Supplementary Figure 5A, B, http://links.lww.com/CM9/C189]. We further checked the DNA methylation status of the 6 cytokines. Our data showed that cg26081812, which is located at TSS1500 on IL-5, was hypomethylated in both DM and PM groups, and three CpG sites (cg24990612, cg18598016, and cg06905823) at the TSS 200 and TSS1500 regions of *PDGFB*, were hypermethylated in both the DM and PM groups [Supplementary Figure 5C, http://links.lww.com/CM9/C189].

In our analysis, we integrated DNA methylation data with gene expression profiles, revealing a notable correlation between DNA methylation and gene expression. We

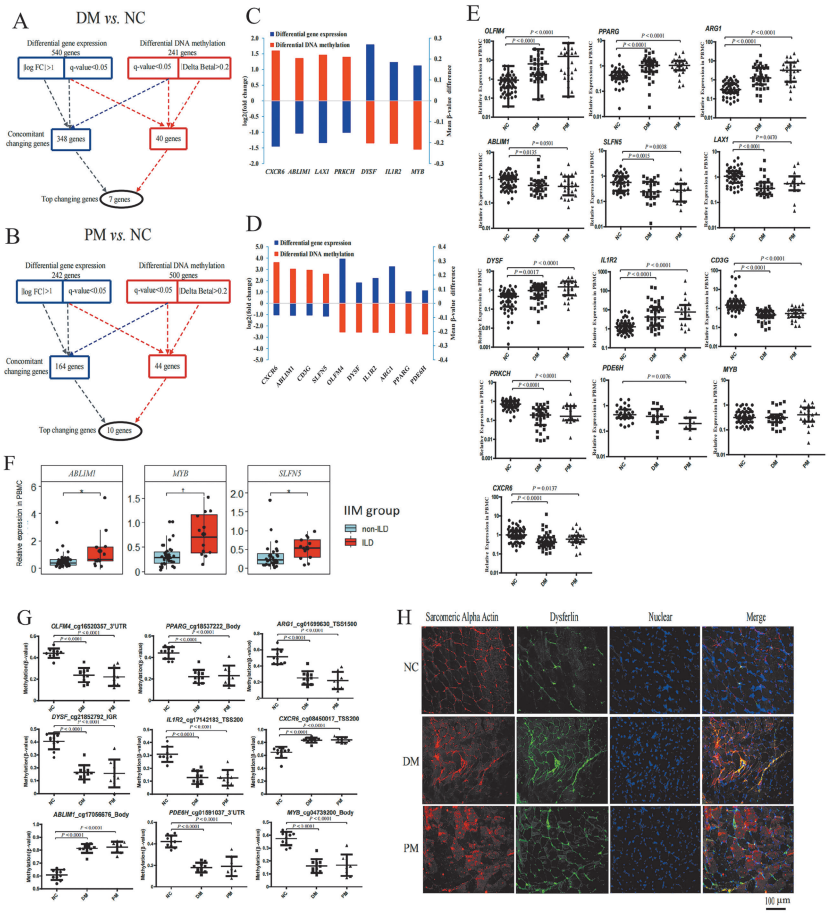

Figure 1: Integrated analysis of DNA methylation and gene expression data in DM and PM patients. (A, B) Pipeline illustrating the selection of the genes with differential methylation and the greatest changes in gene expression in the DM (A) and PM (B) *vs.* NC. The concomitant changing genes were identified based on the criteria of a *q*-value <0.05 and an absolute $\log_2$FC >1 in differentially expressed genes, as well as a *q*-value <0.05 in differentially methylated DNA sites or the criteria of a *q*-value <0.05 in differentially expressed genes and a q-value <0.05 with absolute delta beta >0.2 in differentially methylated DNA sites. The dashed line indicates the selection criteria for these genes. (C, D) The genes with the greatest changes in differential methylation and expression in DM patients (C) and PM patients (D). The blue bars represent gene expression, with the length of each bar corresponding to the $\log_2$(fold change) value in the DM or PM *vs.* NC groups (left Y-axis). The red bars represent DNA methylation, and the length of each red bar represents the value of the mean β-value difference in the DM or PM *vs.* NC groups (right Y-axis). (E) Real-time PCR was used to detect the mRNA expression levels of the top 13 genes. (F) The genes that differed between the IIM ILD and non-ILD groups. (G) Bisulfite-modified DNA pyrosequencing was used to detect the methylation status of the top 9 methylation sites. (H) Immunofluorescence was used to detect the expression level of dysferlin in muscle tissues. DM: Dermatomyositis; DNA meth: DNA methylation; FC: Fold change; Gene exp: Gene expression; IIM: Idiopathic inflammatory myopathies; ILD: Interstitial lung disease; NC: Normal controls; PBMC: Peripheral blood mononuclear cell; PM: Polymyositis.

identified shared methylated CpG sites in DM and PM, such as those in *CXCR6*, *ABLIM1*, *DYSF*, and *IL1R2*, as well as specific methylated genes, underscoring the complexity and specificity of the regulatory mechanisms at play. The dynamic characteristics of DNA methylation profiles, coupled with their responsiveness to disease-associated changes, position them as promising biomarkers in IIMs.

In conclusion, this study emphasizes the crucial roles of DNA methylation in IIMs. This study marks the comprehensive investigation into the impact of methylation on gene transcription as the etiology of DM and PM at the global genome level. This study also highlights the common and specific methylation sites in DM and PM and reveals that *IFN*-related genes and cytokines/chemokines are also regulated by DNA methylation.

Conflicts of interest

None.

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